

Project Initialization and Planning Phase

Date	1 July 2026
Team ID	
Project Title	OptiCrop – Crop Recommendation System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective of OptiCrop is to develop a machine learning-based web application that recommends the most suitable crop based on soil nutrients and environmental conditions. The system helps farmers make informed decisions, improve crop productivity, and reduce the risk of crop failure.
Scope	The project analyzes important agricultural parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall to recommend the most suitable crop. The solution includes a trained machine learning model integrated with a Flask-based web application, allowing users to receive crop recommendations through a simple and user-friendly interface.
Problem Statement	
Description	Farmers often rely on traditional farming practices and personal experience to select crops. Incorrect crop selection due to a lack of scientific analysis of soil nutrients and environmental conditions can lead to low agricultural productivity and financial losses.
Impact	Providing accurate crop recommendations based on soil and environmental parameters helps farmers choose appropriate crops, improve crop yield, reduce unnecessary resource usage, and support sustainable agriculture.
Proposed Solution	
Approach	Develop a machine learning model trained on a crop recommendation dataset to analyze soil nutrients and environmental conditions. Integrate the trained model into a Flask web application that allows users to enter agricultural parameters and receive instant crop

	recommendations.
Key Features	🎬 Machine learning-based crop recommendation. 🎬 Predicts the most suitable crop using seven input parameters: • Nitrogen (N) • Phosphorus (P) • Potassium (K) • Temperature • Humidity • pH • Rainfall 🎬 User-friendly web interface built with Flask. 🎬 Real-time crop prediction. 🎬 Displays a brief description of the recommended crop. 🎬 Responsive design for desktop and mobile devices. 🎬 Deployed online using Render for easy accessib

	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computing Resources	Intel/AMD Processor (Dual Core or above)
Memory	RAM specifications	Minimum 8 GB RAM
Storage	Disk space for data, models, and logs	Minimum 10 GB Free Disk Space
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Joblib, Unicorn
Development Environment	IDE	Visual Studio Code, Jupyter Notebook
Data		
Data	Source, size, format	Kaggle Crop Recommendation Dataset, 2200 records , CSV format